Pump Station Maintenance Manual, Stations PS-3 and PS-4

SYNTHETIC SAMPLE DOCUMENT. Written for the personal-ai-brain evaluation set. The utility, equipment and limits are invented.

1. Scope

This manual covers routine maintenance of the four centrifugal pumps at stations PS-3 and PS-4 of the river intake. Each pump is driven by a 250 kW induction motor at 1,480 rpm and moves up to 1,100 cubic metres of raw water per hour.

Maintenance is planned in three tiers: weekly walk-round inspections by the operator, monthly condition monitoring by the reliability technician, and an annual overhaul by the mechanical contractor. Chapter 2 sets the vibration limits, chapter 3 the bearing lubrication schedule, and chapter 4 the shutdown and isolation procedure.

2. Vibration monitoring

Vibration is measured on both motor bearings and both pump bearings as RMS velocity in mm/s, in the 10 Hz to 1 kHz band, following ISO 10816-3 for machines on rigid foundations.

Limits for these pumps:

Below 2.8 mm/s - good condition, no action.

2.8 to 4.5 mm/s - acceptable, record the trend and check monthly.

4.5 to 7.1 mm/s - alert: increase monitoring to weekly and plan a repair.

Above 7.1 mm/s - danger: stop the pump within 24 hours and inspect.

A rise of more than 25% between two monthly readings is treated as an alert even when the absolute value is still in the acceptable band. Most alerts on these pumps have been caused by bearing wear or by impeller imbalance after debris strikes.

3. Bearing lubrication

Pump bearings are grease-lubricated rolling element bearings. Use a lithium-complex grease of NLGI grade 2. Do not mix with polyurea greases.

Regrease the pump bearings every 2,000 running hours with 30 grams per bearing. Regrease the motor bearings every 4,000 running hours with 20 grams per bearing. Add grease slowly with the pump running and stop when the bearing temperature starts to rise.

Bearing temperature should stay below 80 C. A temperature above 90 C, or a rise of 15 C above the normal running value, means the pump must be stopped and the bearing inspected. Over-greasing is a common cause of high temperature after maintenance.

4. Shutdown and isolation

Before any work inside the pump casing, the pump must be isolated under the permit to work system. Close the suction and discharge valves, lock both with personal padlocks, and lock out the motor isolator at the motor control centre.

Drain the casing through the drain valve and confirm zero pressure on the casing gauge before opening any flange. Two people must be present for confined-space entry into the suction well.

After work, the permit holder removes the locks, the operator primes the pump, and the pump runs for 30 minutes under observation before it returns to automatic control.